Smart Data Analysis in Medical And Health Research

How AI Assistants Can Elevate Your R Programming

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Learning Objectives

> **Note**
>
> **By the end of this session, you will be able to:**

1. **Understand** the Data Analysis Workflow (REAAAPP)
2. **Explore** AI Assistants and their roles in data analysis
3. **Learn** different approaches to integrate AI Assistants with R programming

Tailored to your experience level — beginner, intermediate, or advanced

Biography

Professor, Medical Epidemiologist and Public Health Medicine Specialist & Professor

- Fellow of **the American College of Epidemiology** (FACE)
- Interests in disease modelling: **epidemiological, statistical & AI methods**
- Introduced R in the postgraduate programme in **2014**
- Listed in the **top 2% scientists worldwide** since 2021
- myanalytics.com.my

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Presentation Roadmap

Part 1

REAAAPP Workflow

Systematic data analysis workflow

Part 2

AI Assistants

Overview of AI assistants for R

Part 3

Integration Approaches

Three ways to integrate AI with your R workflow

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Part 1: Data Analysis Workflow

Understanding the REAAAPP Workflow

The REAAAPP Workflow

A systematic approach to data analysis and statistical modelling

R — Reading Data

E — Exploring Data

A — Making Assumptions

A — Applying models and methods

A — Assessing results and models

P — Presenting results and interpretations

P — Publishing results, data, and codes

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AI Powers Each REAAAPP Stage

Stage	How AI Assists
Read	Generate import code for CSV, Excel, SPSS, SQL
Explore	Suggest summaries, visualizations, patterns
Assume	Recommend assumption checks
Apply	Write model code (regression, ML, mixed)
Assess	Interpret diagnostics, suggest improvements
Present	Create Quarto reports, build Shiny apps
Publish	Generate reproducible documents



Key Insight: AI accelerates each stage while *you* retain intellectual oversight

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Part 2: AI Assistants

AI Assistants for R Programming

Before AI Assistants

Traditional Workflow

🔍 **Google Search** — Hunt for code snippets across forums

📚 **Read Books** — Manually adapt examples to your data

💬 **Stack Overflow** — Post questions, wait hours or days

AI-Powered Workflow

⚡ **Describe** your problem in natural language → get **instant R code**

🔄 **Iterate and refine** with follow-up prompts — no context switching

Major AI Assistants for R

Assistant	Key Strengths	Best Use Case
ChatGPT	Broad knowledge, conversational	General R coding & learning
Claude	200K context, artifacts	Complex analysis, code review
Gemini	Google integration, 1M+ context	Multi-modal data analysis
Copilot	IDE integration, real-time	In-editor coding

Each AI assistant has unique capabilities — choose based on your specific needs

Part 3: Integration Approaches

Integrating AI with Your R Workflow

Three Approaches to Integrate AI with R

Choose based on your workflow preferences and technical comfort level

🌐

Browser interface — **easy start, no setup**

📦

ellmer Package

Programmatic access — **automated workflows**



IDE Integration

In-editor AI — **seamless assistance**

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Approach 1: Web-Based Workflow

🌐 Using AI Assistants via Browser

1. Generate prompt with R Prompt Generator
2. Submit to ChatGPT, Claude, or Gemini
3. Share data samples and requirements
4. AI generates complete R solution
5. Copy to RStudio and test
6. Iterate and refine prompts

✅ **Advantages:** Easy to start, no setup, conversational | ⚠️ **Limitations:** Manual copy-paste, context management

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Approach 2: ellmer Package

📦 Using AI Programmatically from R

```r
library(ellmer)
# Set up chat with system prompt
chat <- chat_openai(
  system_prompt = "You are an expert R programmer who prefers tidyverse."
)
# Request R code
response <- chat$chat(
  "Generate R code to:
  1. Read CSV file
  2. Calculate summary statistics by group
  3. Create a ggplot visualization"
)
```

Supports: OpenAI, Claude (Anthropic), Gemini (Google) — consistent API across providers

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Approach 3: IDE Landscape

🖥️ Three Major AI-Integrated IDEs for R

Positron IDE

Positron Assistant
Ask · Edit · Agent
Databot for EDA
MCP support

VS Code

GitHub Copilot
Claude Sonnet 4.6
Claude Code ext.
Continue (OSS)

Google Antigravity

Agent-first IDE
Gemini 3 powered
BigQuery / Cloud Run
Natural language dev

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Positron IDE: Positron Assistant

🟢 Posit's Next-Generation IDE with Integrated AI

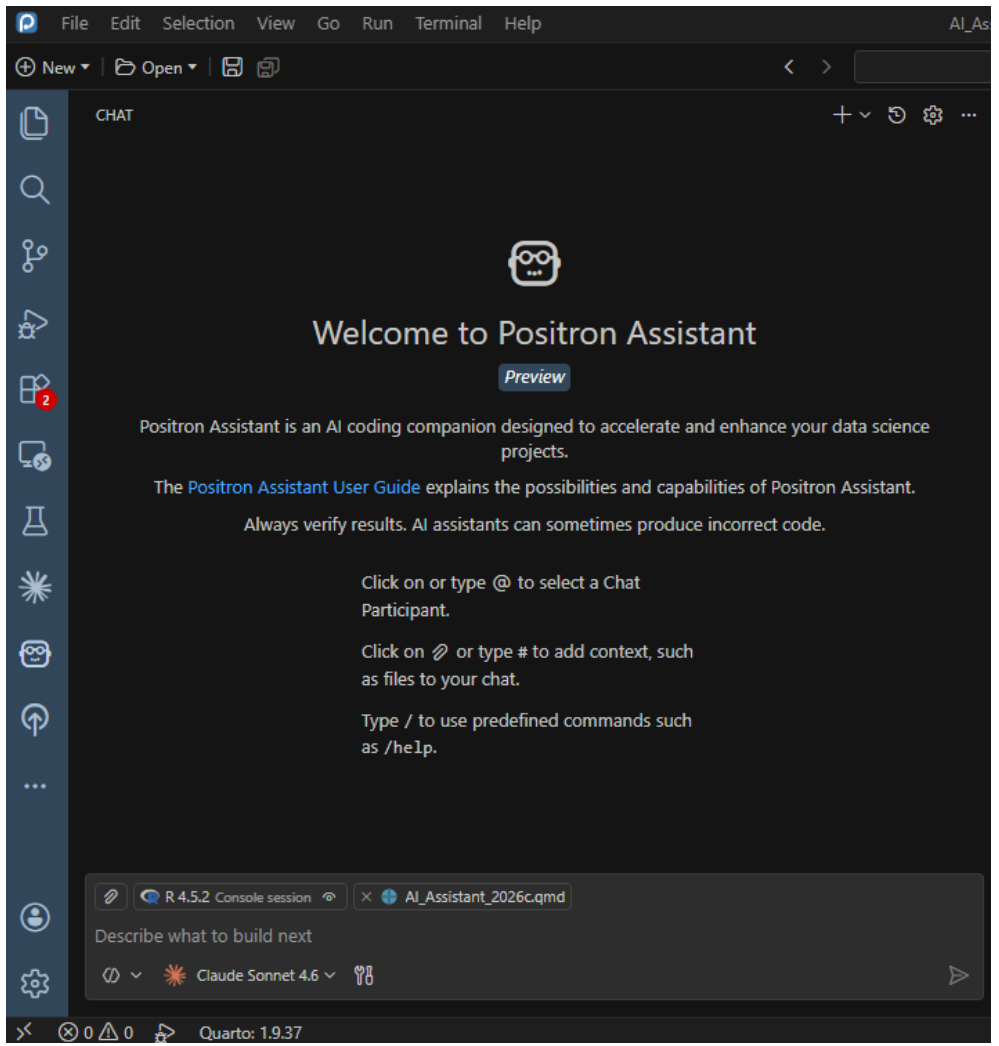
Mode	Function
Ask	💬 Conversational Q&A about code & data
Edit	✏️ Apply AI code changes directly
Agent	🤖 Autonomous execution, multi-step tasks

Multi-model · MCP support · Custom instructions per project



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Positron IDE: Positron Assistant

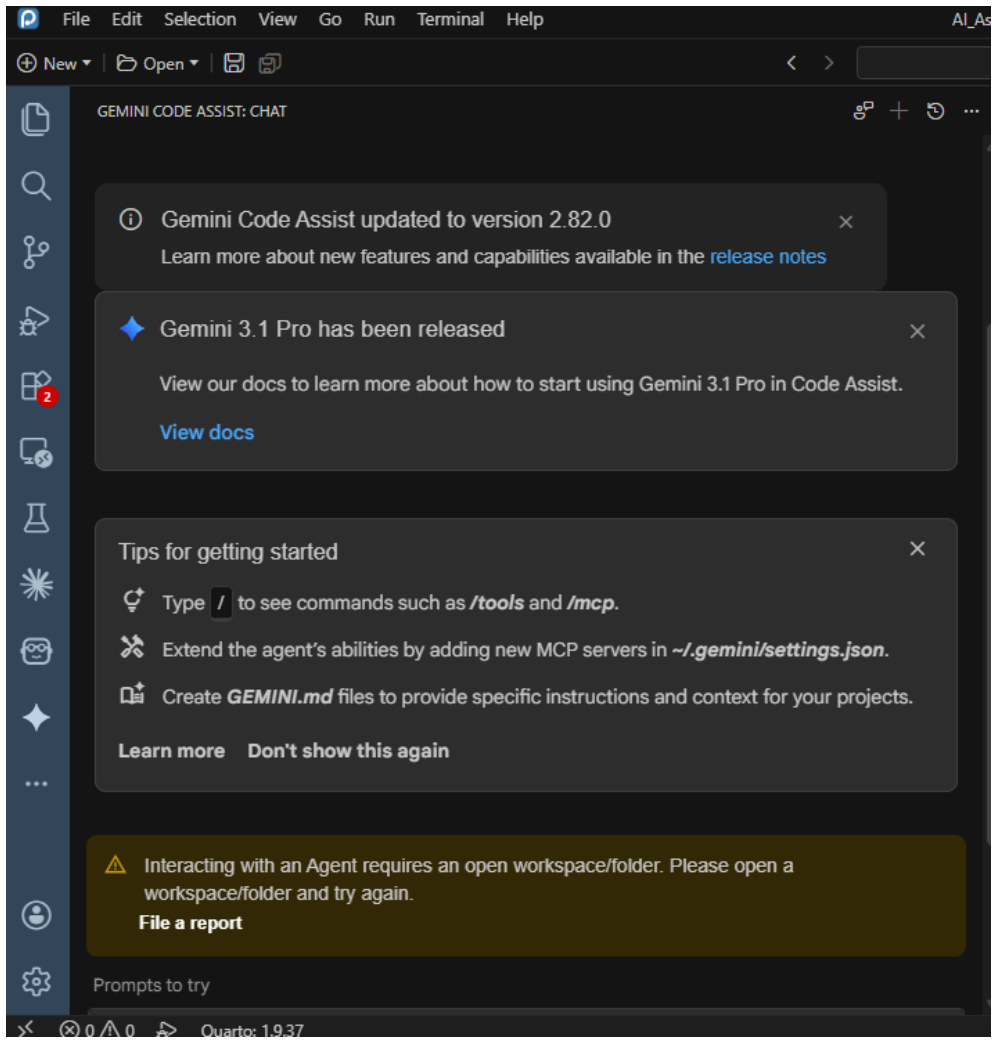


- API key for Anthropic (Claude)
- API key for OpenAI
- GitHub OAuth for Github Copilot Bedrock and Snowflake

Positron Assistant

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Positron IDE: Gemini Code Assist



Gemini Code Assist in Positron IDE

🟢 Google's AI-powered coding companion. Integrated directly into IDEs. It provides inline code completion, multi-modal chat, debugging, and automated testing.

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Installing AI & Productivity Extensions in Positron IDE

🧩 Extend Positron with AI assistants and essential productivity tools

Recommended AI Extensions

🤖 **Claude or GitHub Copilot** *(built-in via Positron Assistant)*

🟣 **Claude Code** — Anthropic's agentic coding extension

🔵 **Gemini Code Assist** — Google's AI coding assistant

Recommended Productivity Extensions

📄 **Quarto** — render `.qmd` documents from the IDE

📑 **PDF Viewer** — preview rendered PDF reports in-pane

📝 **Markdown Preview** — live-render markdown notes & READMEs

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Folder, Subfolder and Files

🧩 Organize files including dataset, resources, R codes and implementation in the suitable folder, sub-folders and files.

Subfolders

🤖 **Dataset** - Positron will read this dataset

🟣 **Resources** — R codes example or main reference papers

Files

📄 **implementation plan** — your workflow written in MD format

📑 **R files** — ask AI to create R codes

Implementation Plan

```
1
2  # Data Analysis Implementation Plan: Ingestion to GLM Output
3
4  Implementation plan for a data analysis project, from raw data ingestion to generating a GL
5
6  ## 1. Project Organization and Data Ingestion
7  * Setup the project directory structure with dedicated subfolders: `data/raw/`, `data/proce
8  * Read the source dataset from `data/raw/data.csv` using `read_csv()`, ensuring blank or "NA
9  * Clean and standardize column headers automatically into consistent snake_case formatting u
10 * Mutate the `treatment` variable into a structured factor with explicitly ordered referenc
11 * Convert the `outcome_status` column into a binary outcome factor variable named `outcome`
12 * Coerce the `age` vector into a standardized numeric format using `as.numeric()`.
13
14 ## 2. Exploratory Data Analysis & Baseline Statistics
15 * Load the prepared dataset from the cached file back into the active workspace as an objec
16 * Isolate and subset the critical analysis variables (`treatment`, `age`, and `outcome`) us
17 * Construct a baseline demographics table (`tbl_base`) stratified across the `treatment` ar
18 * Calculate and append descriptive bivariate p-values across treatment tiers using `add_p()`
19
20 ## 3. Generalized Linear Modeling (GLM) & Presentation Formatting
```

Avoid These Traps

⚠️ During Analysis

- ❌ Your prompts are vague
- ❌ Not providing additional resources
- ❌ Trying to be perfect (overly long prompts)
- ❌ Planning too little

⚠️ After Analysis

- ❌ Not verifying AI output
- ❌ Not validating results
- ❌ Not cleaning generated code
- ❌ Letting AI be your master

💡 **Remember:** Instead of saving time, you'll spend more correcting bad code — *verify everything*

Summary

📋 **REAAAPP Workflow**	**provides structure**
🤖 **AI Assistants**	**accelerates analysis**
🧠 **Your Domain Expertise**	**ensures quality**

✅ **Efficient, Reproducible Data Analysis and in the top 5% most skilled medical data analysts in Malaysia**

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Thank You

🙏 Questions?

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